

App Release Runway

Rev 2 — rewritten 2026-09-06 after the 8.4 upgrade, the Phase 6B patch, and Session 4.3. **Scope narrowed to the app.** OpenEMR is now governed by the owner-authored *OpenEMR 8.4 Master Setup Guide v4.1*, which supersedes this document's former Track A along with the v2.3 deploy guide and the v3.0 billing guide.

Correction to rev 1. Rev 1 recorded the four new-route 401s as an ACL problem alone. That was wrong and it cost an install cycle. OpenEMR derives the required OAuth scope from the route path — `POST .../billing` needs `user/billing.c` — so the routes are gated at **two independent layers**: OAuth scope and ACL. Both had to be satisfied. Tell them apart by the message: ACL says *“Organization policy does not have permit access resource”*; scope says *“Unauthorized”*.

Where things stand

Item	State
OpenEMR 8.4	Done — Upgraded 2026-09-05. Live reports 8.4.0, database 543. Schema ran clean; menus and Connectors held.
OAuth clients	v3 registered with 49 scopes, then v4 with 54 for the 6B routes. Both enabled.
Phase 6B route patch	Done — Built, installed, acceptance 17/17 on the live instance. Adds POST billing (fee-sheet charge), POST order (procedure order + code rows), and GET <code>/api/codes</code> , plus reads, void, and order-status update. Gap 1 closed for charges, orders, and code search.
Sessions 4.1 / 4.2 / 4.4	Done — Merged (PR #19, #22, #29). Workspace, scheduling, clinical completeness.
Session 4.3	Done — Merged (PR #31). Patient clinical read, POA acting gates, case-manager scoped read. Live preflight still owed — the build sandbox could not reach the EMR during the upgrade window, so it was proven against a mock.
Session 4.5	Not started. The last build session before release.

The 6B patch ships as a derived image built from `build.context` in `docker-compose`, so no `docker compose pull` can revert it. The monthly patch routine becomes `docker compose build --pull && docker compose up -d`.

Blocking Session 4.5 — yours, not the app team's

The app team cannot reach the deployed environment. Until these land, 4.5 cannot prove anything live.

Do this	Why
Swap <code>OPENEMR_CLIENT_ID</code> and <code>OPENEMR_CLIENT_SECRET</code> in the deployed environment to the v4 client, then restart	The app is still authenticating as an older client. Without the swap the 6B routes stay unreachable from the running app.
Confirm the app's sync indicator reports the expected scope count	This is the fast proof the swap took effect.
Disable the v2 client and delete the <code>SQNSx...</code> duplicate	An unused enabled client is audit surface.

Do this	Why
Load ICD-10-CM (Administration → Other → External Data Loads)	Does not block the charge write — <code>billing.justify</code> is free text. It does leave <code>GET /api/codes</code> empty, FHIR Condition uncoded, and OpenEMR's own Fee Sheet diagnosis picker blank.
Set the billing NPI, and each clinician's NPI on their user record	Signing is blocked without them, by design.

Session 4.5 — the last build session

Prompt still to be written. Five pieces:

- 1. Native 8.4 writes.** Prescriptions move to `POST /api/prescription` (verified working, surfaces in `MedicationRequest`). Re-enable the vitals write — the endpoint is fixed and now produces both a row and a FHIR `Observation`. Encounter PUT works, but only when user and group are in the body.
- 2. The 6B routes.** Charges to `POST billing`, orders to `POST order`, code search to `GET /api/codes`.
- 3. Retire the workarounds — in a separate commit.** The app-side `clinical_orders` and `encounter_billing` stores and the `[GFC ORDERS] / [GFC CODING]` note blocks. Add the native path, prove it live, then remove the workaround. Never both in one commit.
- 4. Per-visit billing facility picker.** Writes `form_encounter.billing_facility`. Note it is **not** a charge field — `addBilling()` has no such parameter and the billing table no such column.
- 5. Run Session 4.3's live preflight** as part of 4.5's live round trip, since 4.3 was proven only against a mock.

Contract for the build: Part VII of the Master Setup Guide v4.1. Acceptance: its Appendix A.

Still open on the server — known, not blocking

None of these stop Tuesday. Each has a working app-side accommodation already in place.

Item	Accommodation
Document route keyed by numeric pid; read-back unproven. FHIR <code>DocumentReference</code> and Coverage still 403	Serve the care-plan PDF from the Drive reference on <code>client.carePlanDocs</code> . Org-level read grant is Phase 8.6.
ICD-10 not loaded — FHIR Condition still uncoded	Coding assist fills a problem's code from the app's own prior coded encounter for that problem uuid.
No appointment update route on 8.4	Reschedule and cancel continue to use the tombstone swap from 4.2.
Encounter rows still returned twice	The FHIR bundle flattener dedupes by <code>resourceType</code> and id.

Before Bethel touches it

Walk one full visit end to end on test data: chart, follow-up note, diagnosis, prescription, order, service code, sign and close, then confirm the charge is visible in Billing Manager with the **correct CPT** — not the diagnosis code.

Why that last check is spelled out. Phase 6B acceptance took three runs. Both earlier failures were ours, not the server's, and the second was introduced by the fix for the first — a loop reused the variable holding the CPT, so the charge billed the diagnosis instead. Neither failed loudly: the row looked right in Billing Manager and would have surfaced weeks later as a denial. Both sites now share `normalizedDiagnoses()`, and acceptance asserts stored values rather than status codes. **Assert what landed, not what returned 200.**

Then re-document any test encounters created before Session 4.4 — their notes were orphaned by the defect fixed in PR #29 and cannot be signed.

Document map

Document	Owns
GFC_OpenEMR_84_Master_Setup_Guide_v4_1.pdf	OpenEMR entirely — identity, facilities, codes, billing, compliance. Part VII is the app-alignment contract; Appendix A is acceptance. Supersedes v2.3, v3.0, and this document's former Track A.
This document	The app sequence to release.
docs/openemr-patches/8.4.0-p1/	The 6B route patch: Dockerfile, route wrapper, scopes registration, acceptance script.
OPENEMR_SERVER_DEFECTS_2026-08.md	Live preflight findings and the scope-versus-ACL distinction.
CLAUDE.md	Running status. Authoritative on what is merged.